Fundamentals of Fog and Edge Computing

BCSE313L

WINTER 2024 - 2025

Module 3 - Orchestration of Network Slices in Fog, Edge, and Clouds

Introduction

- Digital transformation is reshaping various sectors through innovative applications such as smart cities, vehicle-to-vehicle (V2V) communication, virtual reality (VR), augmented reality (AR), and remote medical surgery.
- These applications have **distinct connectivity** and **performance requirements**, presenting significant challenges for traditional network infrastructures that rely on a one-size-fits-all approach.

Digital Transformation Has Led Explosion in

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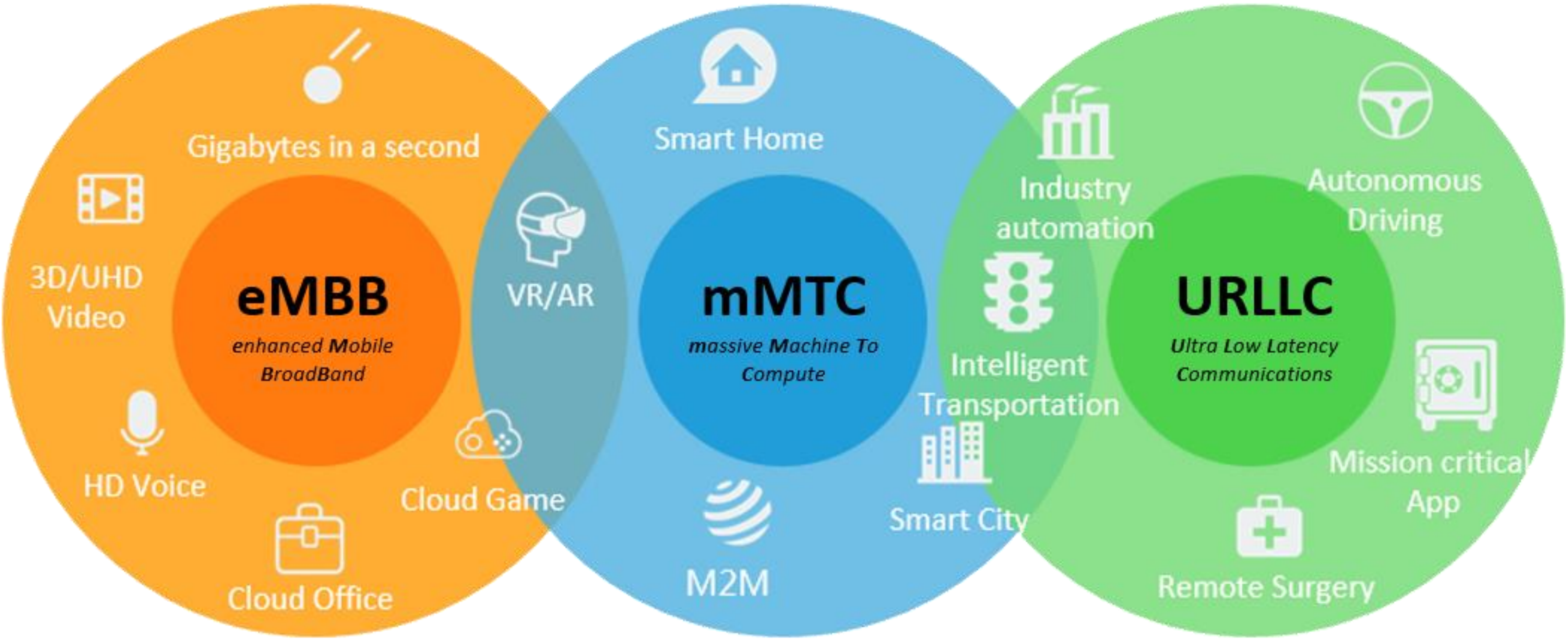
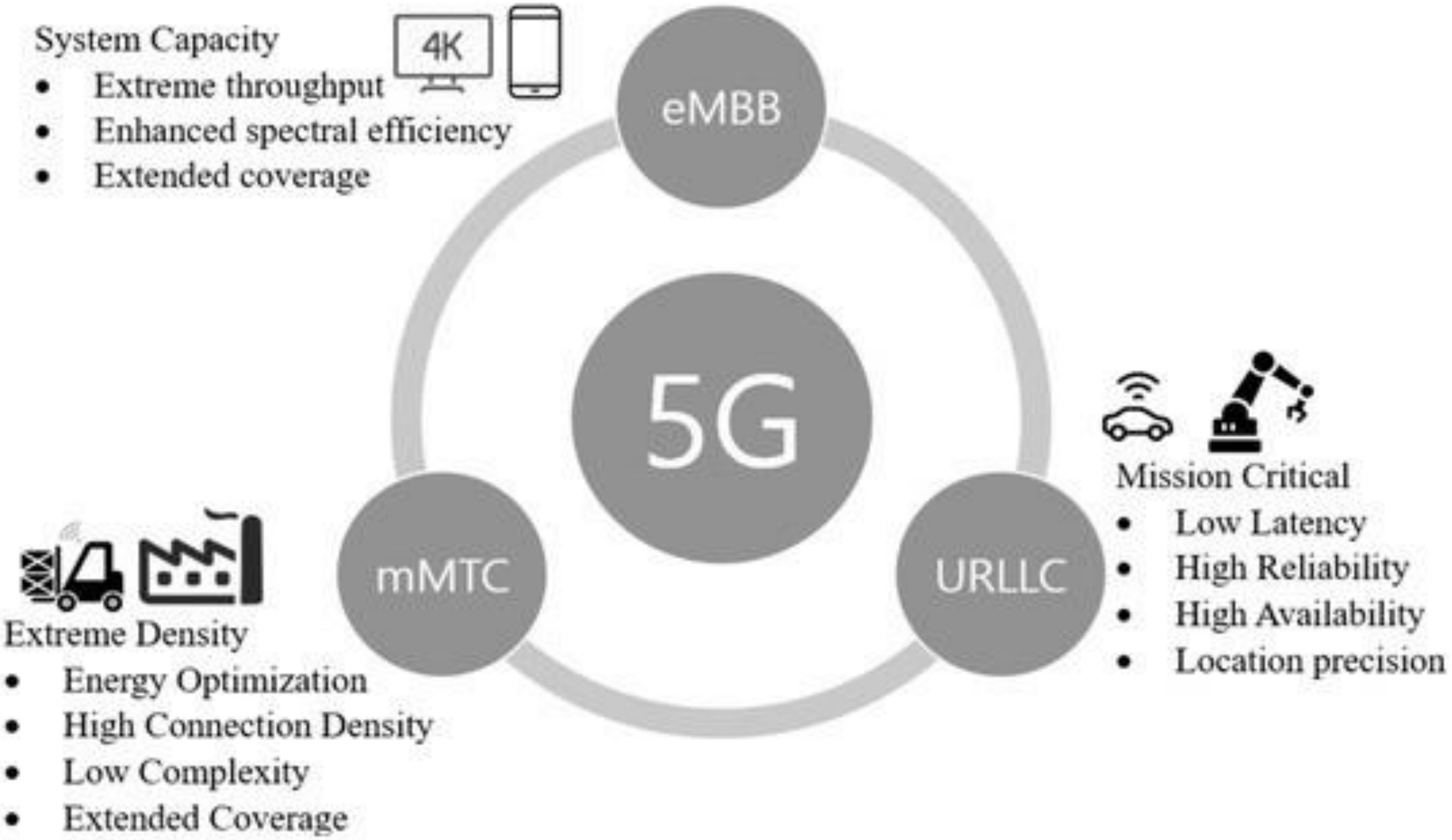


Introduction - Key Use Cases and Quality of Service Requirements



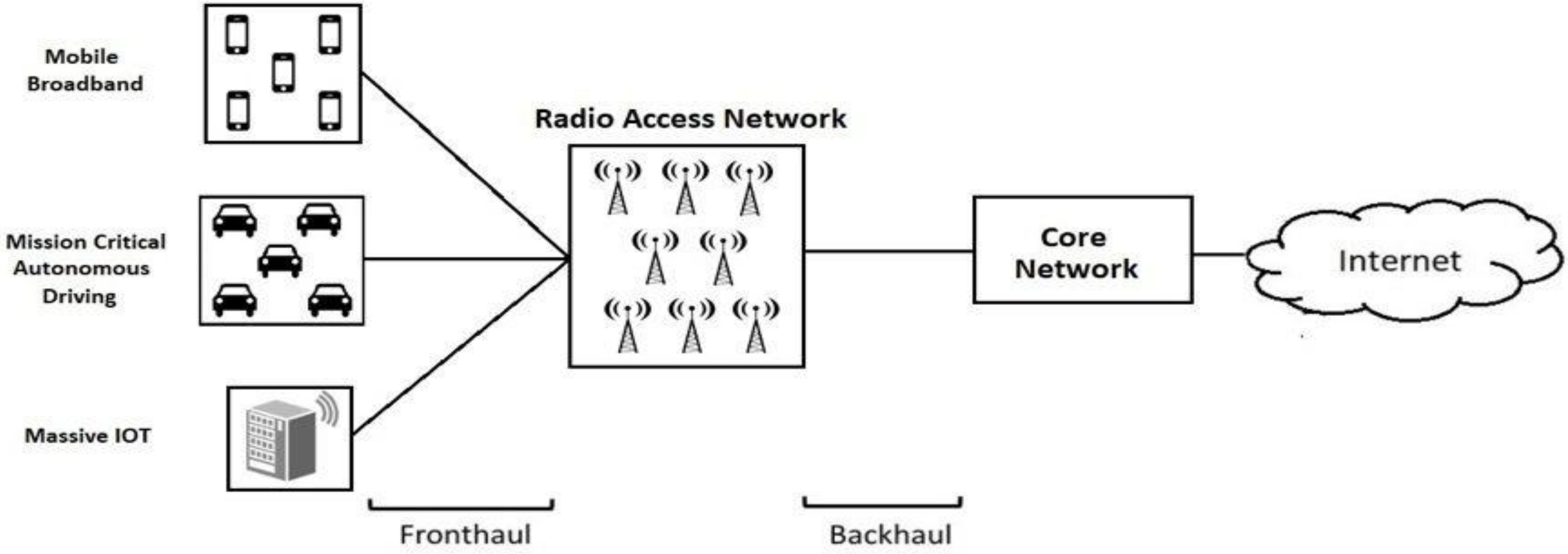
Introduction

- The 5G infrastructure public-private partnership (5G-PPP) has categorized these applications into three main use case families:

  - Enhanced mobile broadband (eMBB),
  - Massive machine-type communications (mMTC)
  - Ultra-reliable low-latency communication (uRLLC)

Introduction – Traditional Network Architecture Limitations

- The diverse quality of service (QoS) requirements of these use cases highlight the limitations of traditional network architectures, which cannot efficiently support the varying demands of different vertical services.



Introduction – Traditional Network Architecture Limitations

- Manual Configurations
- Limited Scalability
- Inconsistent Quality of Service (QoS)
- Complexity in Management
- Rigid Architecture
- Vendor Dependence

Introduction – Traditional Network Architecture Limitations

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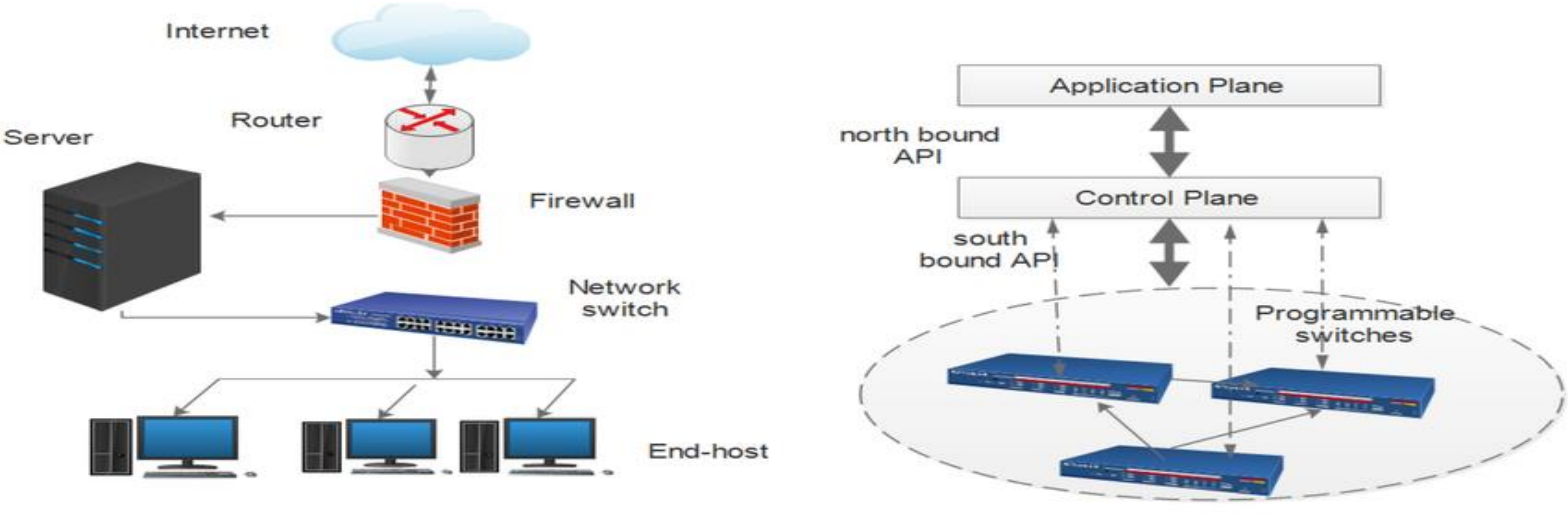


a Traditional Network architecture

b SDN architecture

Introduction – Traditional Network Architecture Limitations

Manual Configuration:

- Traditional networks rely on manual configuration of individual devices such as routers and switches. This process is time-consuming, increases operational complexity, and is prone to human error, making it difficult to maintain consistency across the network

Limited Scalability:

- These architectures struggle to scale effectively with the increasing number of devices and bandwidth demands. They often require significant reconfiguration when new devices are added, which can lead to oversubscription and performance issues

Introduction – Traditional Network Architecture Limitations

Inconsistent Quality of Service (QoS):

Implementing consistent QoS policies across numerous devices is challenging due to the manual nature of configurations. This inconsistency can lead to inadequate performance for applications requiring high reliability and low latency, such as uRLLC

Complexity in Management

The complexity of managing traditional networks increases with the number of protocols and devices involved. This complexity makes it difficult to troubleshoot issues that affect multiple devices, leading to longer resolution times and increased downtime

Introduction – Traditional Network Architecture Limitations

Rigid Architecture:

Traditional networks have a hierarchical architecture that is not easily adaptable to changing business needs or traffic patterns. This rigidity prevents dynamic resource allocation, which is essential for accommodating varying demands from different applications

Vendor Dependence:

Many traditional networks are built on proprietary hardware and software solutions, which can limit flexibility and increase costs when integrating new technologies or scaling services

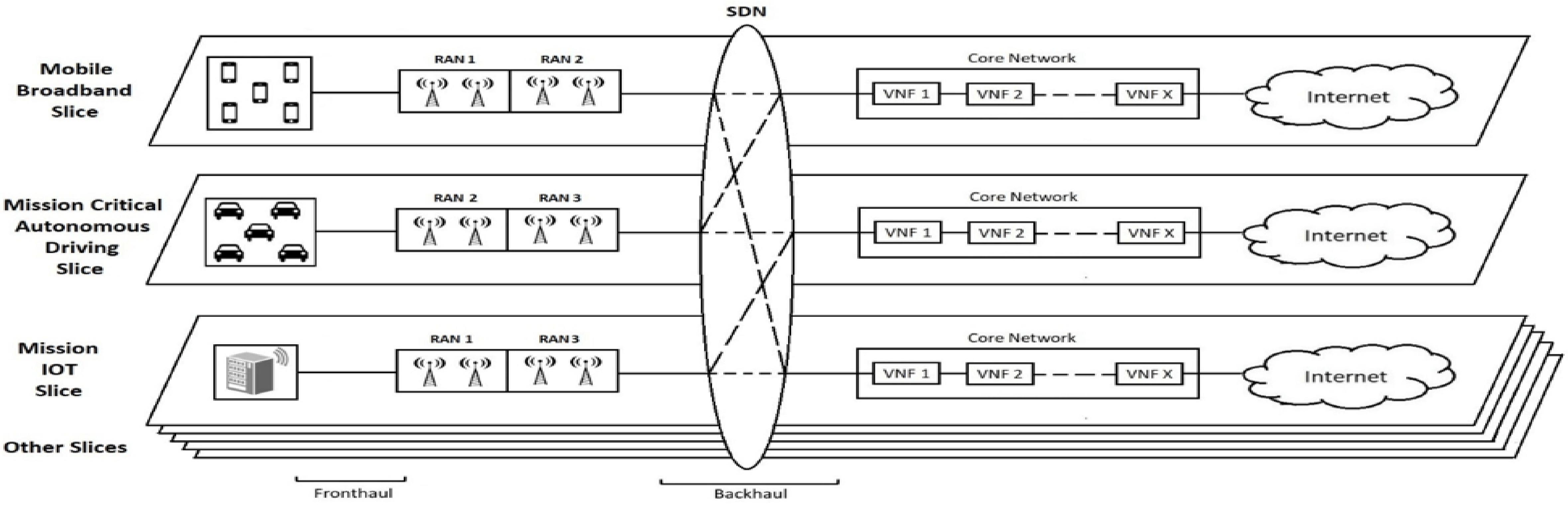
Network Slicing - as a Solution

- A promising approach to address these challenges is network slicing, which partitions a physical network into multiple isolated logical networks.

- This concept mirrors server virtualization in cloud computing, allowing for dynamic resource allocation based on specific application needs.

Network Slicing - as a Solution

- Network slicing enables the creation of tailored virtual networks that can adapt to the unique requirements of different services, effectively providing a network-as-a-service (NaaS) model



Network Slicing - Benefits

Cost Efficiency

- Reduces the need for dedicated physical infrastructure by allowing multiple services to share the same resources

Flexibility

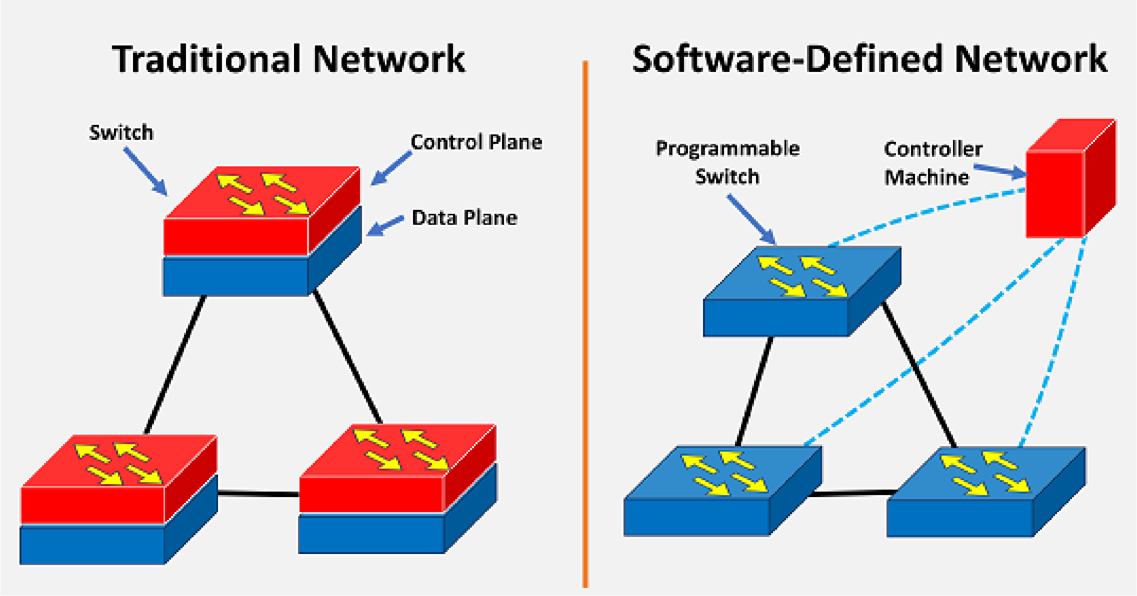
- Operators can quickly deploy and manage network slices tailored to specific customer needs.

Scalability

- Easily scales resources up or down based on demand, accommodating varying service requirements

Role of SDN and NFV in Network Slicing

- Software-Defined Networking (SDN) and Network Function Virtualization (NFV)/Network virtualized function (NVF) are foundational technologies that enhance network slicing capabilities:
  - SDN separates control and data planes, providing centralized management and improved visibility across the network.
  - NFV shifts network functions from hardware to software applications on general-purpose hardware, increasing elasticity and reducing costs



Network Slicing in 5G

- Network slicing in 5G refers to the **creation of multiple virtual networks on top of a single physical network**.

- Each slice operates independently with its own resources, topology, traffic flow, management policies, and protocols, allowing for tailored connectivity for specific applications or services

Network Slicing in 5G

- The primary goal of network slicing is to meet the **unique requirements of various use cases**—ranging from enhanced mobile broadband (eMBB) for high-bandwidth applications to ultra-reliable low-latency communication (uRLLC) for critical services like remote surgery

Network Slicing in 5G - Challenges

Resource Provisioning

- Managing resource allocation among multiple slices is complex due to varying resource affinities that can change over time

Mobility Management

- Handling mobility across slices can complicate the management of resources and connectivity

End-to-End Orchestration

- Coordinating the orchestration and management of slices across different domains (like RAN and core networks) adds complexity to implementation

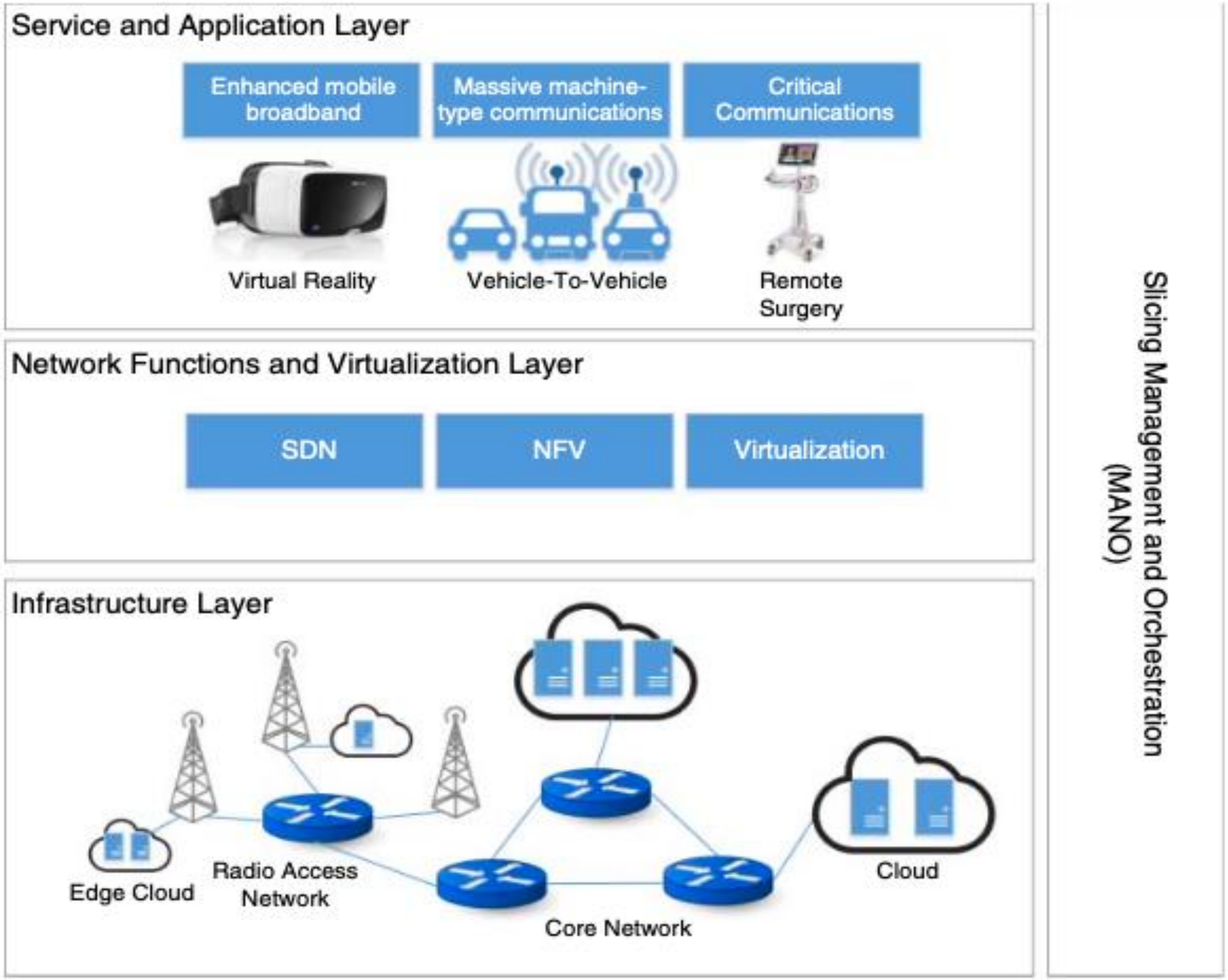
Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**: This layer includes the physical resources such as servers, storage, and networking equipment.

- **Network Function Layer:** This layer encompasses the virtualized network functions that can be deployed within each slice.

- **Service Layer**: This layer focuses on the specific services offered to end-users through the various slices

Network Slicing in 5G – Generic Framework



Network Slicing in 5G – Generic Framework

Use Case
(Connected Devices, Applications, Services, Specific Utility Expectations)

Mapping Functions
(SLAs, Mapping N/w Functions to NVF to Physical resource)

Service and Application Layer

Lifecycle Management
(Managing NVFs, Optimal Placements of NVFs in physical resources)

Chaining of Slices
(Make Slices interact, leverage combined functionalities as needed)

Technical Aspects
(Coarse grained and Fine grained NVFs managing, SDN implementation)

Network Function and Virtualization Layer

Physical Network Architecture
(Remote Cloud, Edge Cloud, Radio Access Network, Cloud-RAN)

Resource Abstraction
(Software Defined Technique, Virtual Storage, Network, Compute, Dynamic allocation)

Policy Implementation
(Policies for deploy, control, manage, Orchestrate for allocation)

Resource Allocation
(Allocation of abstracted resources to the network slices)

Infrastructure Layer

Create Virtual instances from Physical Resource
Establishment of Network Slices

Maps N/W Functions to Virtualized Instance
Construct Service Chains
Allocation of Resources & Functions

Mange Communication between Layers
Lifecycle Management
Adapting and Scaling to needs

Slicing Management and Orchestration (MANO)

Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**
  - It defines the actual physical network topology that supports multiple virtual networks.
  - ***Physical Network Architecture***: The infrastructure layer extends from edge cloud environments to remote cloud setups, encompassing both the radio access network (RAN) and the core network. This layered approach allows for efficient resource allocation and management across various network components .

Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**
  - ***Resource Abstraction***: Utilizing software-defined techniques, this layer facilitates resource abstraction, allowing for dynamic allocation of compute, storage, bandwidth, and other resources to different network slices. This abstraction is critical for enabling flexibility and responsiveness to varying service demands .

Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**
  - ***Policy Implementation***: Within the infrastructure layer, various policies are established to deploy, control, manage, and orchestrate the underlying physical infrastructure. These policies ensure that resources are allocated effectively to meet the specific needs of each slice while maintaining overall network performance .
  - **End-to-End Resource Allocation**: The infrastructure layer plays a crucial role in allocating resources to network slices in a manner that supports upper layers. This allocation is done contextually, ensuring that each slice can operate independently with the necessary resources to fulfill its requirements

Network Slicing in 5G – Generic Framework

➢ Network Function and Virtualization Layer

- ➢ The Network Function and Virtualization Layer operates above the infrastructure layer and is responsible for managing the lifecycle of virtual resources and network functions
- ➢ ***Lifecycle Management***: This layer executes all necessary operations for managing virtualized network functions (VNFs), ensuring optimal placement of these functions within the available physical resources.

Network Slicing in 5G – Generic Framework

- **Network Function and Virtualization Layer**
  - *Chaining of Slices*: It facilitates the chaining of multiple slices to meet specific service requirements. By coordinating how different slices interact, this layer ensures that applications can leverage combined functionalities across slices .
  - *Technical Aspects*: Key technologies such as Software-Defined Networking (SDN) and Network Function Virtualization (NFV) are integral to this layer. They enable efficient management of both coarse-grained and fine-grained network functions across the core and local RAN .

Network Slicing in 5G – Generic Framework

- **Service and Application Layer**
  - The Service and Application Layer represents the topmost layer where various applications and services operate:
  - ***Use Cases***: This layer includes connected devices such as vehicles, IoT devices, mobile phones, and VR appliances. Each application or service has specific utility expectations from the underlying networking infrastructure.
  - ***Mapping Functions***: Based on high-level service requirements, virtualized network functions are mapped to physical resources to ensure compliance with Service Level Agreements (SLAs). This mapping ensures that each application receives the necessary resources without violating performance guarantees .

Network Slicing in 5G – Generic Framework

- **Slicing Management and Orchestration (MANO) in 5G**
  - This layer oversees the functionality of the underlying infrastructure, network functions, and service layers by performing three main tasks:

  - ***Creation of Virtual Network Instances:*** The MANO layer is responsible for creating virtual network instances based on the physical network resources defined in the infrastructure layer. This process allows for the establishment of independent slices that can cater to various applications and services.

Network Slicing in 5G – Generic Framework

- **Slicing Management and Orchestration (MANO) in 5G**
  - ***Mapping Network Functions:*** It maps network functions to these virtualized network instances, facilitating the construction of service chains. This mapping is essential for ensuring that specific service requirements are met through the appropriate allocation of resources and functionalities from the network function and virtualization layer.

Network Slicing in 5G – Generic Framework

- **Slicing Management and Orchestration (MANO) in 5G**
  - ***Lifecycle Management:*** The MANO layer maintains communication between the service/application layer and the network slicing framework. This function involves managing the lifecycle of virtual network instances and dynamically adapting or scaling virtualized resources in response to changing contexts or demands

Network Slicing in 5G – Generic Framework

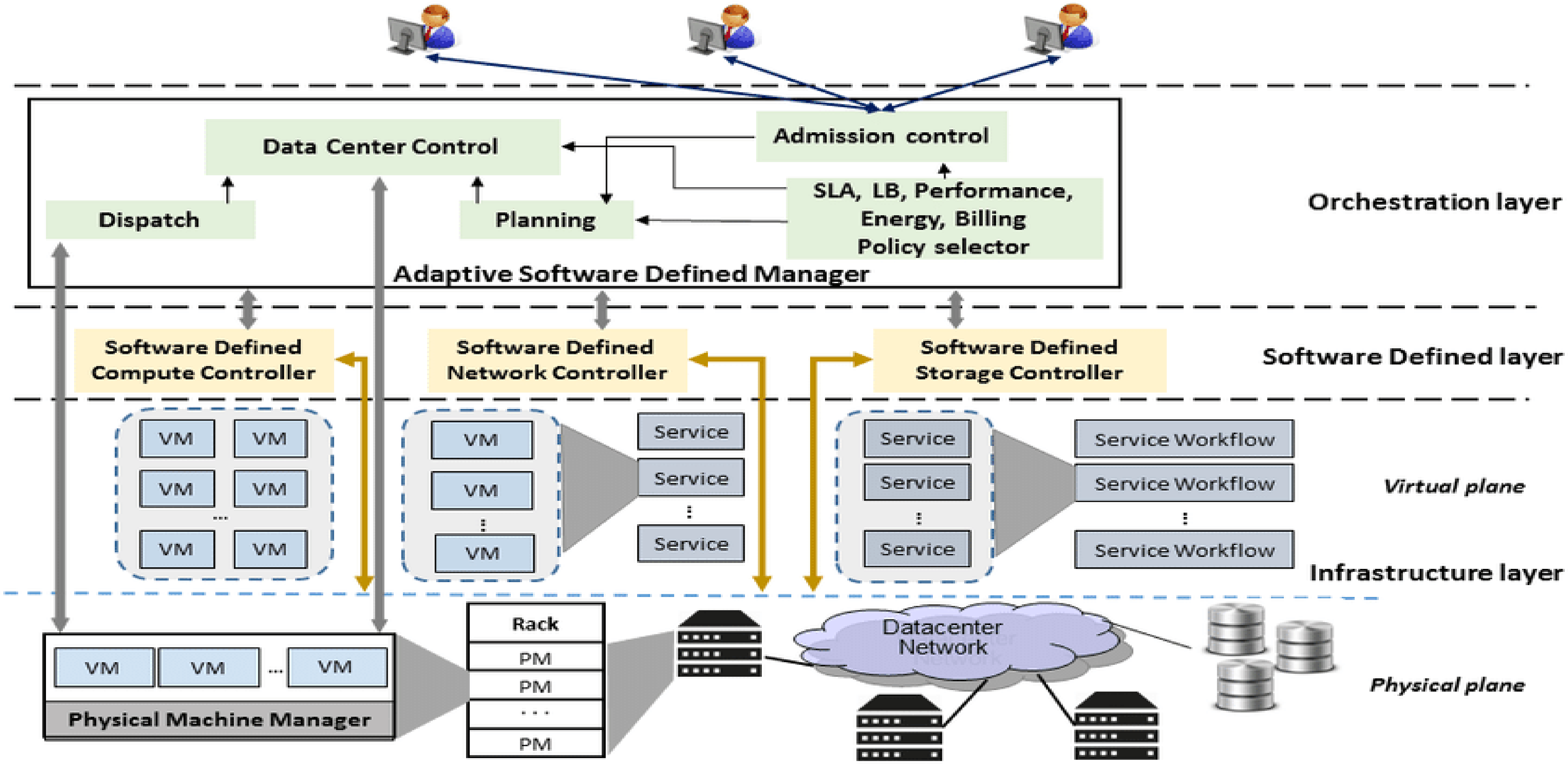
Characteristics of Cloud-Native Network Architecture for 5G		
	Cloud Data Center-Based Architecture:	It provides a logically independent network slicing capability over a cloud data center infrastructure, supporting various application scenarios.
	Cloud-RAN Implementation:	The architecture employs Cloud-RAN to build radio access networks (RAN), facilitating many connections and enabling on-demand deployments of RAN functions as required by 5G.
	Simplified Core Network Architecture:	The core network design is streamlined, allowing for on-demand configuration of network functions through user and control plane separation, unified database management, and component-based functionalities.
	Automated Network Slicing Services:	The architecture implements network slicing services automatically to reduce operational expenses, enhancing efficiency in resource utilization .

Network Slicing in Software Defined Clouds

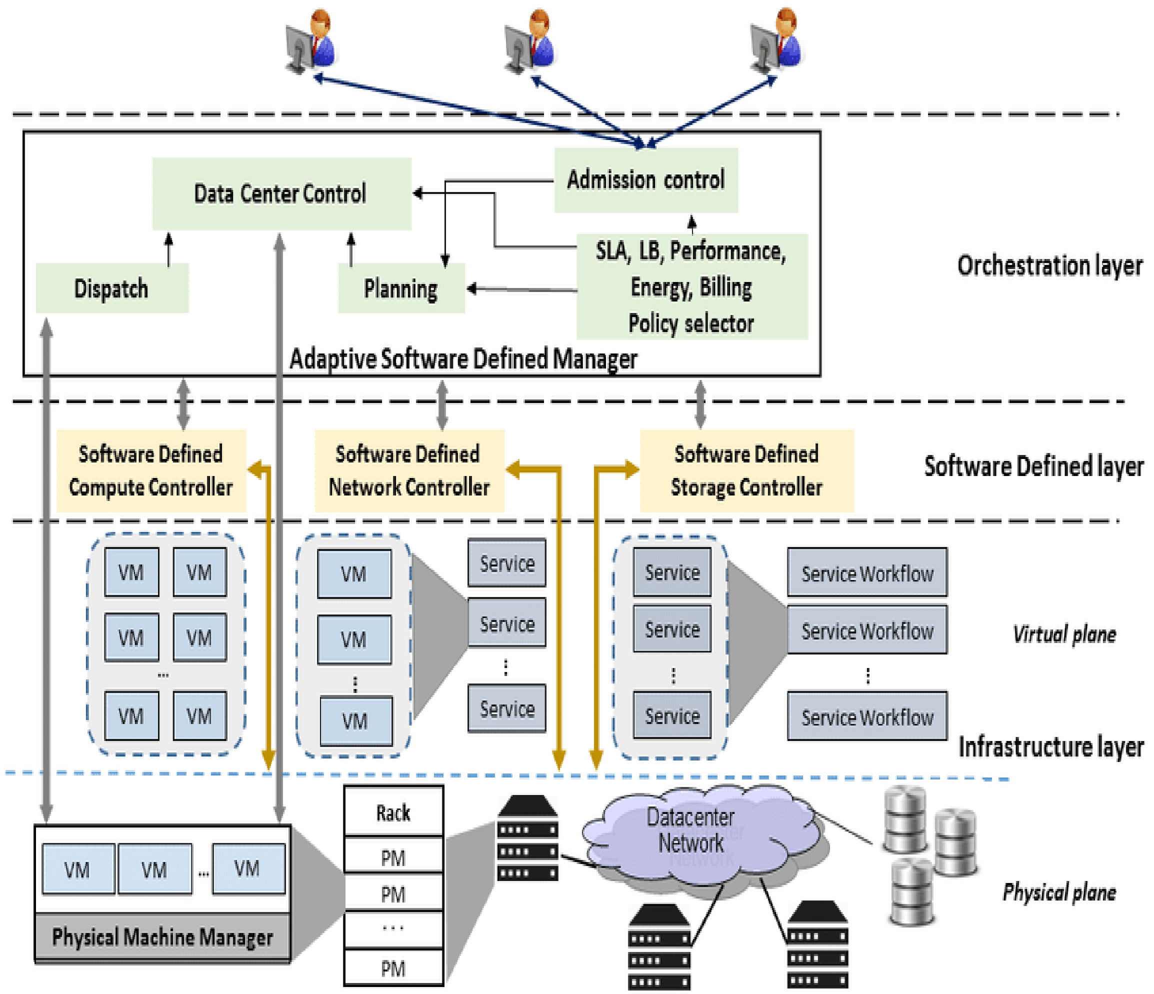
- Virtualization technology has been crucial for resource management and optimization in cloud data centers for the last decade.

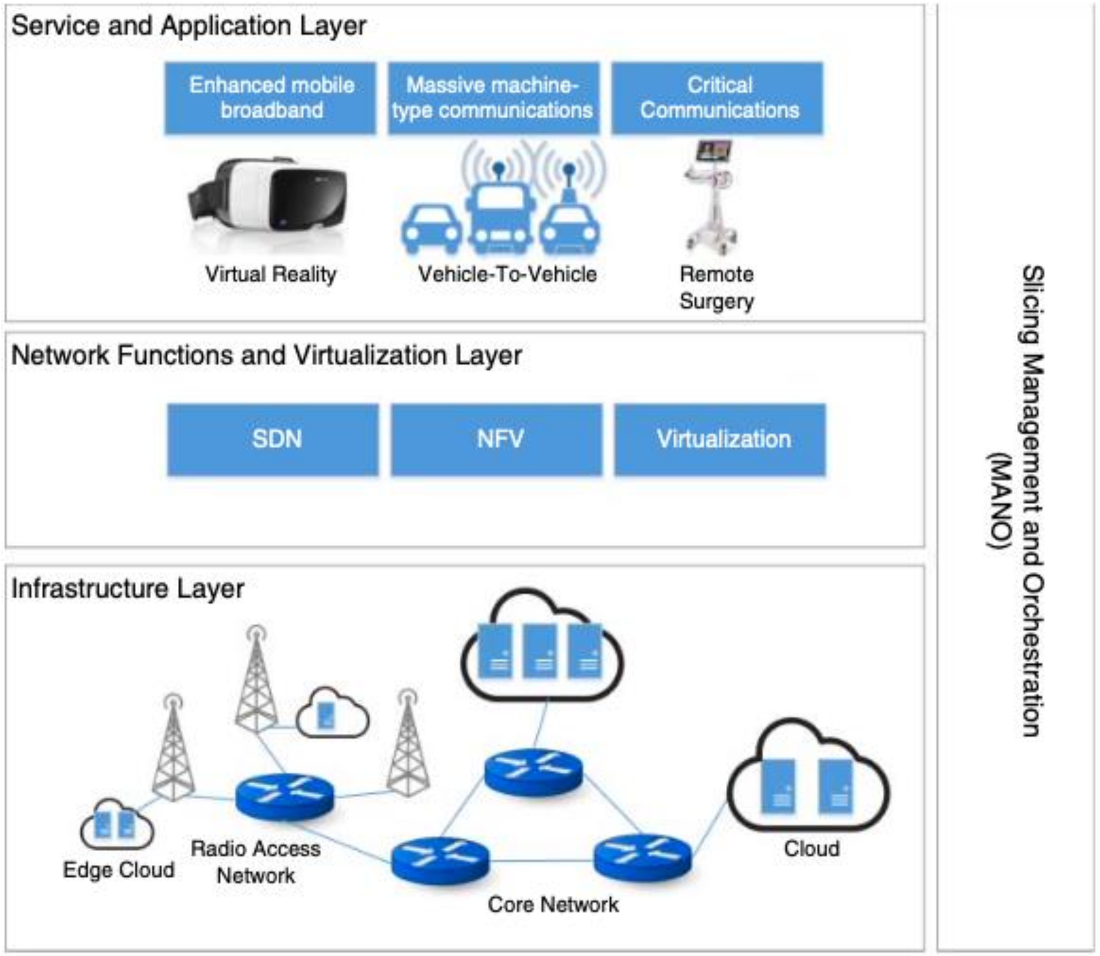
- Research has focused on VM placement and virtual machine (VM) migration to improve utilization and efficiency of physical and virtual servers.

Software Defined Clouds Architecture



Software Defined Clouds Architecture

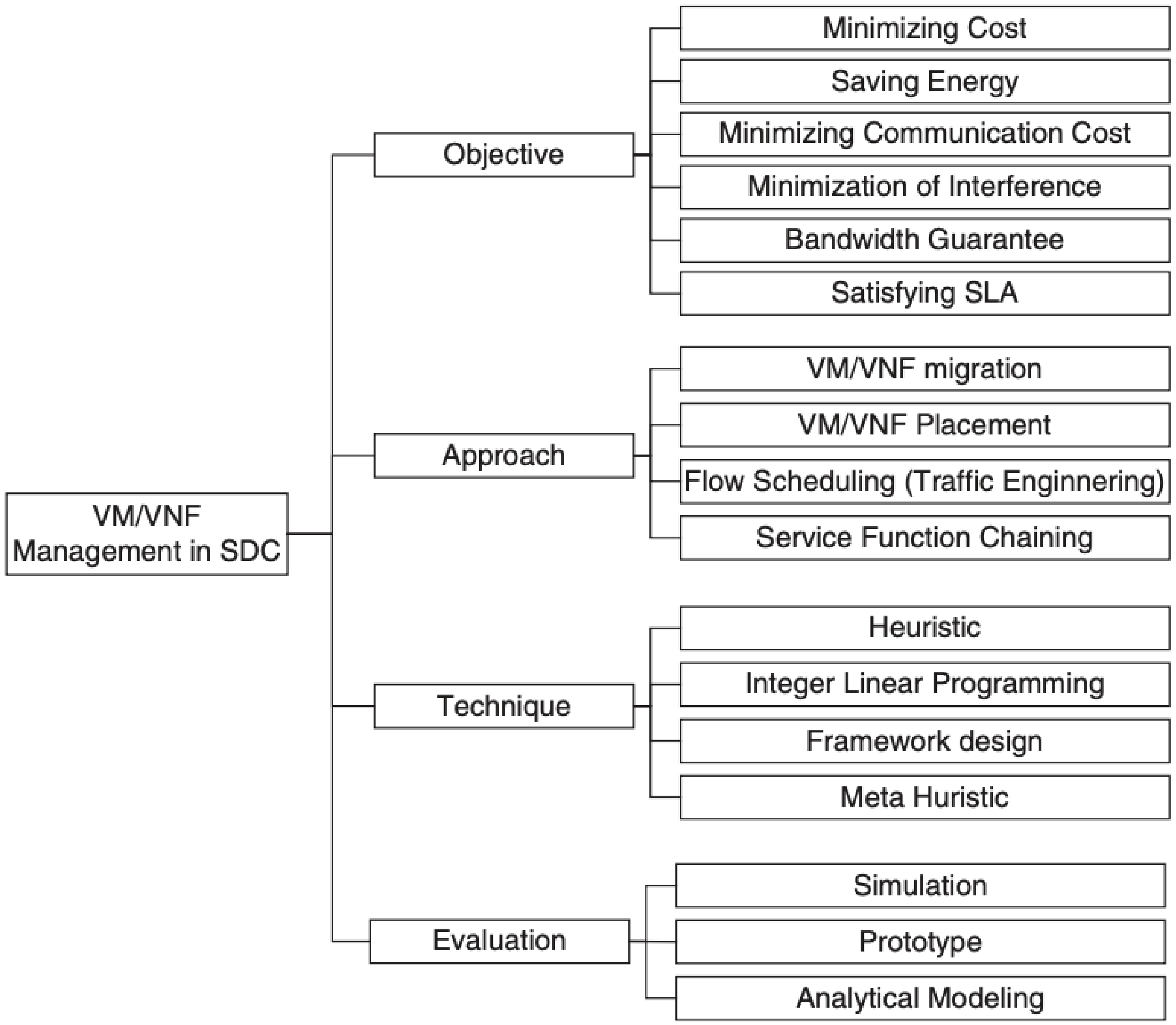




Perspectives on Network Slicing

- It refers to different viewpoints or approaches to understanding and managing network slicing in cloud data centers.
  - **Network-aware VM management:** This perspective focuses on managing virtual machines (VMs) in a way that takes into account the network resources and requirements.
  - **Network-aware VM migration**: This perspective focuses on migrating VMs between different physical or virtual servers while considering network constraints and optimization.
  - **VNF management**: This perspective focuses on managing Virtual Network Functions (VNFs), which are software implementations of network functions, such as firewalls or routers.

Network-Aware VM/VNF Management Taxonomy



Network-Aware VM/VNF Management Taxonomy

Project	Objectives	Approach/Technique	Evaluation
Cziva et al. [29]	Minimization of the network communication cost	VM migration – Framework Design	Prototype
Wang et al. [30]	Reducing the number of hops between communicating VMs and network power consumption	VM placement – Heuristic	Simulation
Remedy [31]	Removing congestion in the network	VM migration – Framework Design	Simulation
Jiang et al. [32]	Minimization of the network communication cost	VM Placement and Migration – Heuristic (Markov approximation)	Simulation
VMPlanner [33]	Reducing network power consumption	VM placement and traffic flow routing - Heuristic	Simulation
PLAN [35]	Minimization of the network communication cost while meeting network policy requirements	VM Placement - Heuristic	Prototype/ Simulation

Network-Aware Virtual Machine Migration Planning

Project	Objectives	Approach/Technique	Evaluation
Bari et al. [37]	Finding sequence of migrations while migration time is minimized	VM migration – Heuristic	Simulation
Ghorbani et al. [38]	Finding sequence of migrations while imposing bandwidth guarantees	VM migration – Heuristic	Simulation
Li et al. [39]	Finding sequence of migrations and destination hosts to balance the load	VM migration – Heuristic	Simulation
iAware [40]	Minimization of migration and co-location interference among VMs	VM migration – Heuristic	Prototype/ Simulation

Virtual Network Functions Management

Project	Objectives	Approach/Technique
VNF-P	Handling burst in network services demand while minimizing the number of servers	Resource allocation - Integer linear programming (ILP)
Cloud4NFV	Providing network function as a service	Service provisioning – Framework design
vConductor	Virtual network services provisioning and management	Service provisioning – Framework design
MORSA	Multi objective placement of virtual services	Placement - Multi-objective genetic algorithm
TVM	Reducing number of hops in service chain	VNF migration - heuristic
SOVWin	Increasing user requests acceptance rate and minimization of SLA violation	VNF placement - heuristic
Clayman et al.	Providing automatic placement of the virtual nodes	VNF placement - heuristic
T-NOVA	Building a marketplace for VNF	Marketplace – framework design
UNIFY	Automated, dynamic service creation and service function chaining	Service provisioning– framework design

Network Slicing Management in Edge and Fog

- Fog computing extends cloud computing to the edge of the network, providing real-time and low-latency processing.

- **Key challenges**: Integrating fog/edge computing with SDN/NFV for efficient management and orchestration of network services.

Network Slicing Management in Edge and Fog

- State of Art Solutions Proposed:
  - Lingen et al. [52]: Proposed a model-driven and service-centric architecture for integrating NFV, fog, and 5G/MEC.
  - Truong et al. [53]: Proposed an SDN-based architecture to support fog computing, with use cases in vehicular adhoc networks (VANETs).
  - Bruschi et al. [54]: Proposed a network slicing scheme for supporting multidomain fog/cloud services using SDN-based network slicing.
  - Choi et al. [55]: Proposed a fog operating system architecture called FogOS for IoT services, addressing scalability, inter-networking, dynamics, and diversity challenges.
  - Diro et al. [56]: Proposed a mixed SDN and fog architecture for prioritizing critical network flows while ensuring fairness among other flows.

Network Slicing Management in Edge and Fog

- **Model-driven and service-centric architecture**: Addresses technical challenges of integrating NFV, fog, and 5G/MEC.
- **Open architecture**: Based on NFV MANO (ETSI) and aligned with OpenFog Consortium (OFC) reference architecture.
- **Two-layer abstraction model**: Integrates cloud, network, and fog with IoT-specific modules and enhanced NFV MANO architecture.
- **SDN-based architecture**: Supports fog computing with identified required components and specified roles.
- **Central network view**: SDN controller manages resources and services, optimizes migration and replication.

Network Slicing Management in Edge and Fog

- Use Cases
  - **Smart Cities**: Fog computing can be used for real-time processing of sensor data from smart city infrastructure, such as traffic management and public safety.
  - **Industrial Automation**: Fog computing can be used for real-time control and monitoring of industrial automation systems, such as manufacturing and process control.
  - **Vehicular Networks**: Fog computing can be used for real-time processing of data from vehicular networks, such as traffic management and safety applications.
  - **IoT Applications**: Fog computing can be used for real-time processing of data from IoT devices, such as smart homes and wearables.

Network Slicing Management in Edge and Fog

- Network slicing scheme: Supports multidomain fog/cloud services using SDN-based network slicing.
- Overlay network: Built using non-overlapping OpenFlow rules for geographically distributed Internet services.
- Experimental results: Significant reduction in unicast forwarding rules compared to fully meshed and OpenStack cases.

Network Slicing Management in Edge and Fog

- **Fog operating system architecture**: Designed for IoT services, addressing four main challenges:
  - Scalability
  - Complex inter-networking
  - Dynamic topology and QoS requirements
  - Diversity and heterogeneity in communications and computing

Network Slicing Management in Edge and Fog

- Four main components:
  - Service and device abstraction
  - Resource management
  - Application management
  - Edge resource management

Network Slicing Management in Edge and Fog

- **Mixed SDN and fog architecture**: Prioritizes critical network flows while ensuring fairness among other flows.

- **QoS requirements**: Satisfies packet delay, lost packets, and maximized throughput for heterogeneous IoT applications.

- **Results**: Efficient serving of critical and urgent flows, with fair allocation of network slices to other flow classes.